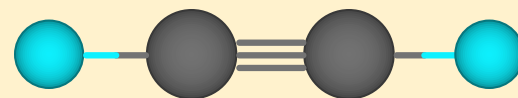
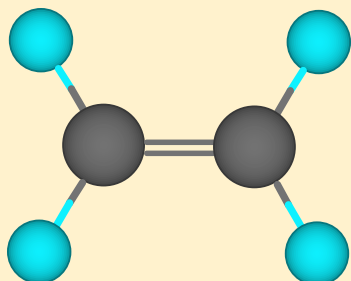
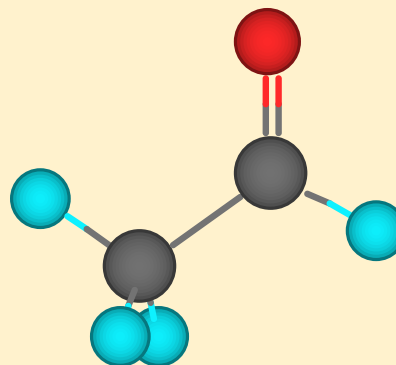
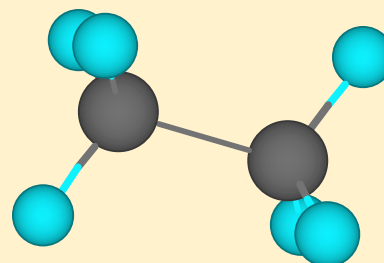
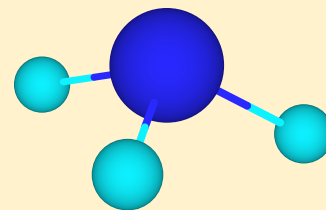
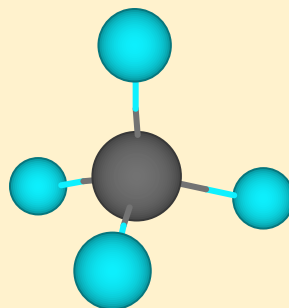


Chapter 1: Bonding, Structure and Reactivity



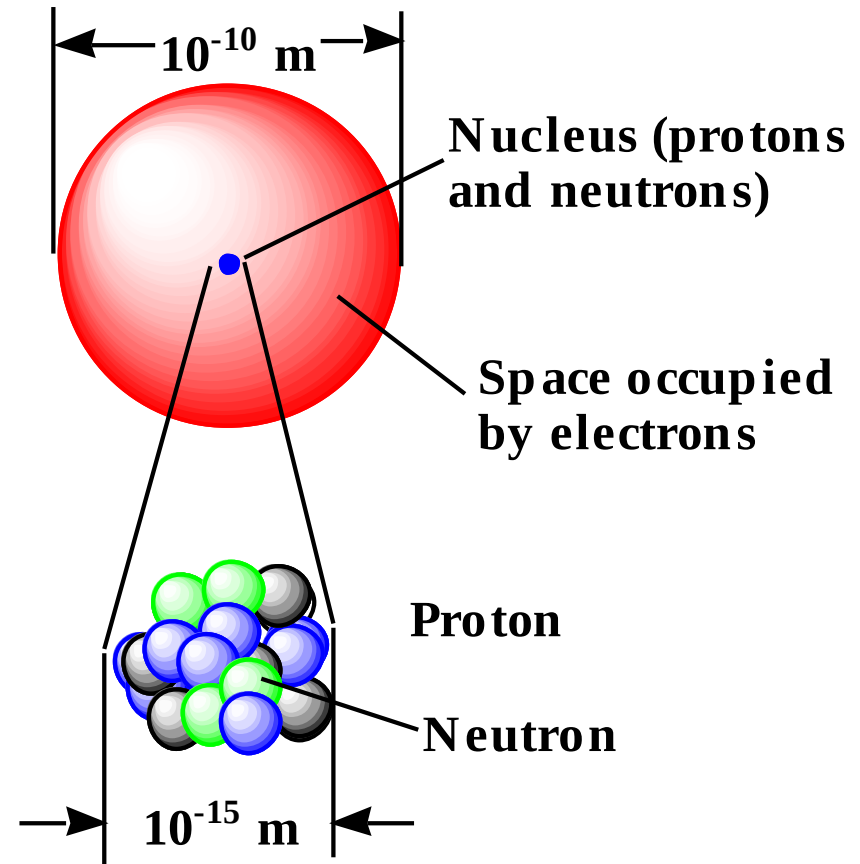
Organic Chemistry

- The study of the compounds of carbon
- Over 10 million organic compounds have been identified
 - about 1000 new ones are discovered or synthesized and identified each day!
 - Carbon atom is a central element
- C is a small atom
 - it forms single, double, and triple bonds
 - it is intermediate in electronegativity (2.5)
 - it forms strong bonds with C, H, O, N, and some metals

Energy Levels and Atomic Orbitals

Electronic Structure of Atoms

- **Structure of atoms**
 - a small dense nucleus, diameter 10^{-14} - 10^{-15} m, which contains positively charged protons, neutrons and most of the mass of the atom
 - extranuclear space, diameter 10^{-10} m, which contains negatively charged electrons



Electronic Structure of Atoms

- Electrons are confined to regions of space called principle energy levels (shells)
 - each shell can hold $2n^2$ electrons ($n = 1, 2, 3, 4, \dots$)

Shell	Number of Electrons Shell Can Hold	Relative Energies of Electrons in These Shells
4	32	higher
3	18	
2	8	
1	2	
		lower

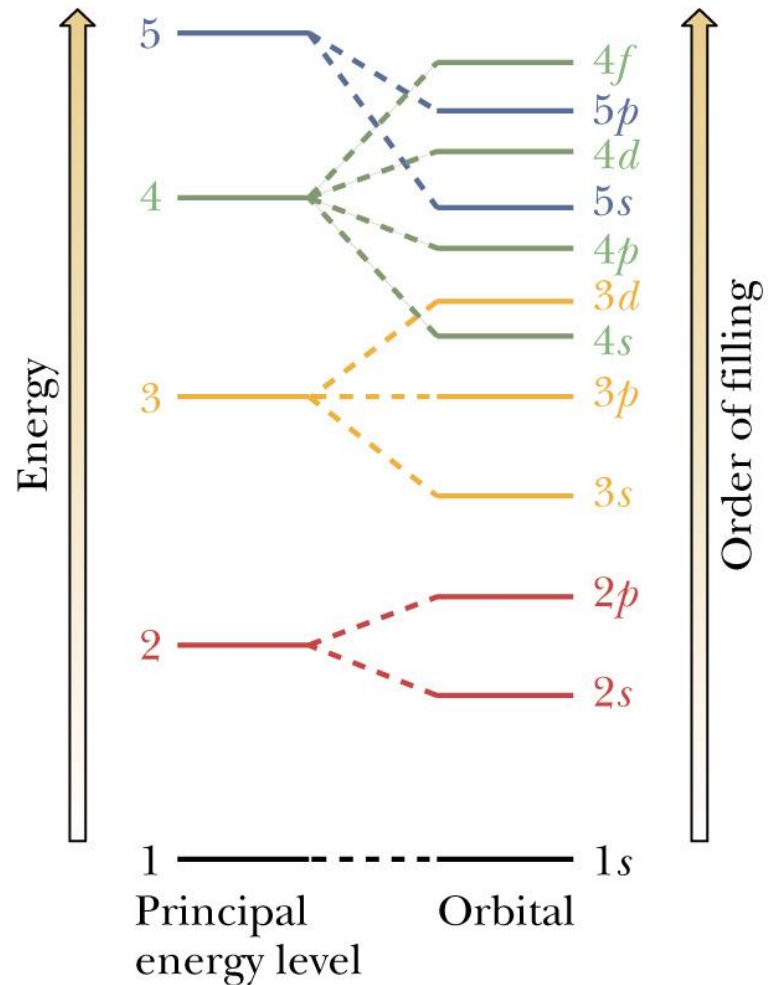
Electronic Structure of Atoms

- Shells are divided into subshells called orbitals, which are designated by the letters *s*, *p*, *d*,.....
 - *s* (one per shell)
 - *p* (set of three per shell 2 and higher)
 - *d* (set of five per shell 3 and higher)

Shell	Orbitals Contained in That Shell
3	$3s$, $3p_x$, $3p_y$, $3p_z$, plus five $3d$ orbitals
2	$2s$, $2p_x$, $2p_y$, $2p_z$
1	$1s$

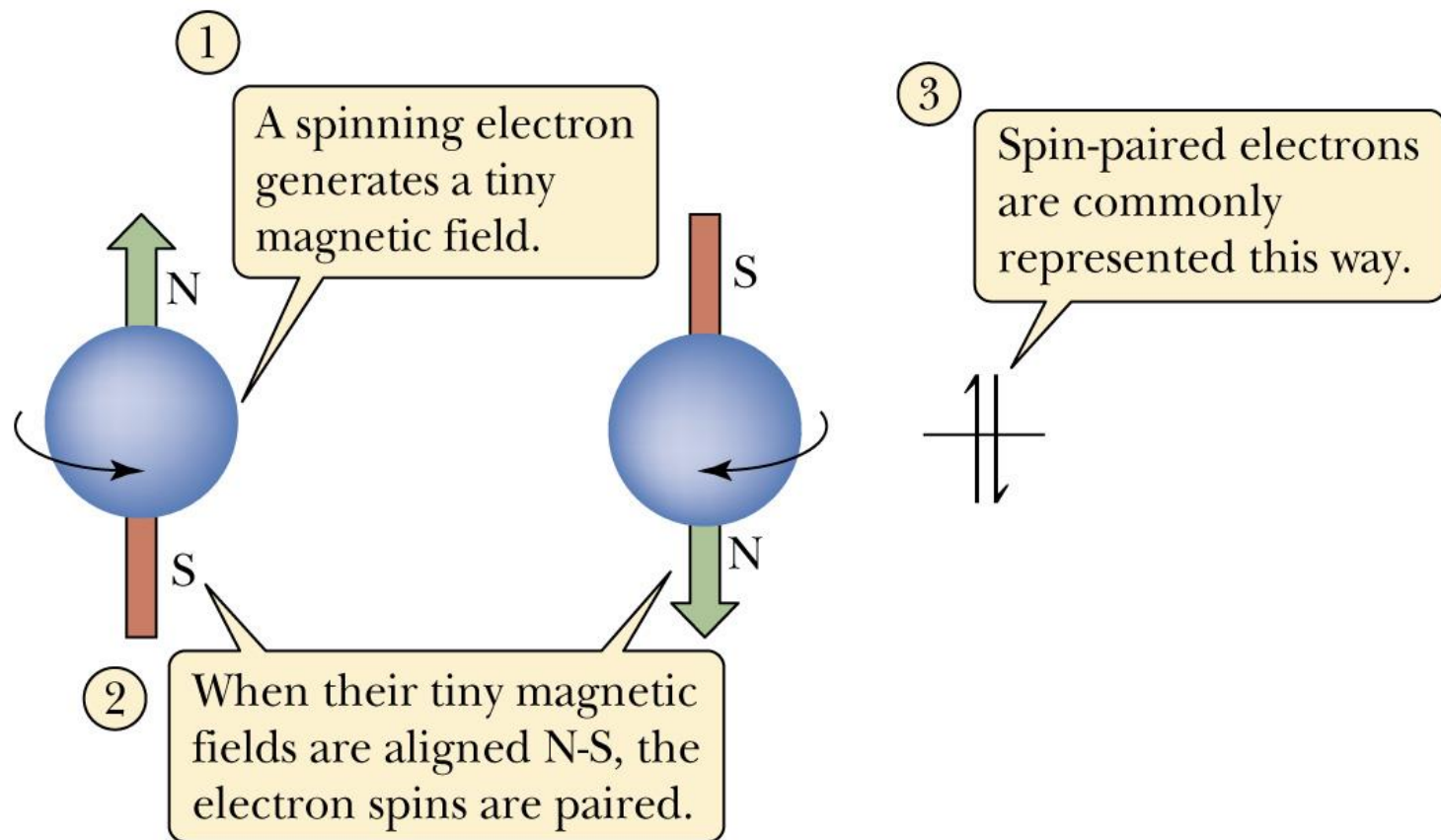
Electronic Structure of Atoms

- **Rule 1:** orbitals fill from lowest energy to highest energy
- **Rule 2:** only two electrons per orbital, spins must be paired
- **Rule 3:** for a set of orbitals with the same energy, add one electron in each before a second is added in any one



Electronic Structure of Atoms

- The pairing of electron spins



Lewis Structures

- **Gilbert N. Lewis**
- **Valence shell:** the outermost electron shell of an atom
- **Valence electrons:** electrons in the valence shell of an atom; these electrons are used in forming chemical bonds
- **Lewis structure**
 - the symbol of the atom represents the nucleus and all inner shell electrons
 - dots represent valence electrons

Lewis Structures

- Lewis structures for elements 1-18 of the Periodic Table

1 A	2 A	3 A	4 A	5 A	6 A	7 A	8 A
H·							He:
Li·	Be:	B·	·C·	·N·	·O·	·F·	·Ne·
Na·	Mg:	Al·	·Si·	·P·	·S·	·Cl·	·Ar·

Bonding In Carbon Compounds:

Lewis Model of Bonding

- Atoms bond together so that each atom in the bond acquires the electron configuration of the noble gas nearest it in atomic number
 - an atom that gains electrons becomes an **anion (negatively charged)**
 - an atom that loses electrons becomes a **cation (+vely charged)**
 - **Ionic bond:** a chemical bond resulting from the electrostatic attraction of an **anion** and a **cation**
 - **Covalent bond:** a chemical bond resulting from two atoms **sharing one or more pairs of electrons**
- We classify chemical bonds as ionic, polar covalent, and nonpolar covalent based on the difference in **electronegativity** between the atoms

Electronegativity

- **Electronegativity:** a measure of the force of an atom's attraction for the electrons it shares in a chemical bond with another atom
- **Pauling scale**
 - increases from left to right within a period
 - increases from bottom to top in a group



Electronegativity

- Electronegativity of atoms (Pauling scale)

TABLE 1.5 Electronegativity Values for Some Atoms (Pauling Scale)

1A		2A												3A		4A	5A	6A	7A
H 2.1																			
Li 1.0	Be 1.5											B 2.0	C 2.5	N 3.0	O 3.5	F 4.0			
Na 0.9	Mg 1.2	3B	4B	5B	6B	7B	8B				1B	2B	Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0		
K 0.8	Ca 1.0	Sc 1.3	Ti 1.5	V 1.6	Cr 1.6	Mn 1.5	Fe 1.8	Co 1.8	Ni 1.8	Cu 1.9	Zn 1.6	Ga 1.6	Ge 1.8	As 2.0	Se 2.4	Br 2.8			
Rb 0.8	Sr 1.0	Y 1.2	Zr 1.4	Nb 1.6	Mo 1.8	Tc 1.9	Ru 2.2	Rh 2.2	Pd 2.2	Ag 1.9	Cd 1.7	In 1.7	Sn 1.8	Sb 1.9	Te 2.1	I 2.5			
Cs 0.7	Ba 0.9	La 1.1	Hf 1.3	Ta 1.5	W 1.7	Re 1.9	Os 2.2	Ir 2.2	Pt 2.2	Au 2.4	Hg 1.9	Tl 1.8	Pb 1.8	Bi 1.9	Po 2.0	At 2.2			

<1.0	1.5 – 1.9	2.5 – 2.9
1.0 – 1.4	2.0 – 2.4	3.0 – 4.0

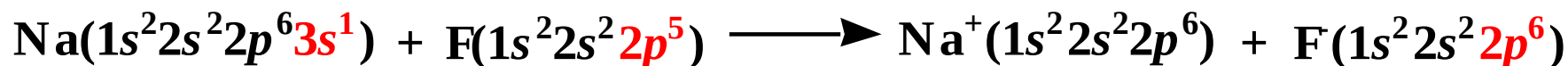
Electronegativity

- Electronegativity and chemical bonding

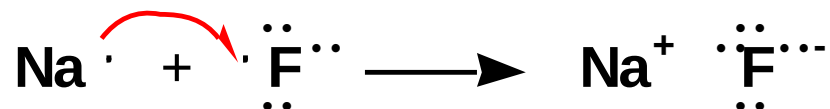
Difference in Electronegativity Between Bonded Atoms	Type of Bond
less than 0.5	nonpolar covalent
0.5 to 1.9	polar covalent
greater than 1.9	ionic

Ionic Bonds

- An ionic bond forms by the transfer of electrons from the valence shell of an atom of lower electronegativity to the valence shell of an atom of higher electronegativity.



- we show the transfer of a single electron by a single-headed (barbed) curved arrow



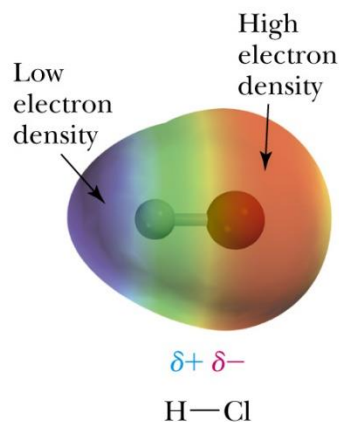
Covalent Bonds

- A covalent bond forms when electron pairs are shared between two atoms whose difference in electronegativity is 1.9 or less
 - an example is the formation of a covalent bond between two hydrogen atoms
 - the shared pair of electrons completes the valence shell of each hydrogen.



Polar Covalent Bonds

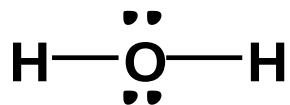
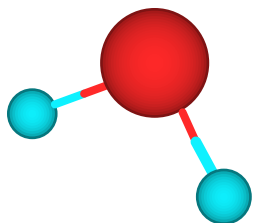
- In a polar covalent bond
 - the more electronegative atom has a partial negative charge, indicated by the symbol δ^-
 - the less electronegative atom has a partial positive charge, indicated by the symbol δ^+
- in an electron density model
 - Red (larger) indicates a region of high electron density
 - blue indicates a region of low electron density



Drawing Lewis Structures

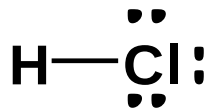
- **To draw a Lewis structure:**
 - **determine the number of valence electrons in the molecule or ion**
 - **determine the arrangement of atoms**
 - **connect the atoms by single bonds**
 - **arrange the remaining electrons so that each atom has a complete valence shell**
 - **show bonding electrons as a single bond**
 - **show nonbonding electrons as a pair of dots**
 - **atoms share 1 pair of electrons in a single bond, 2 pairs in a double bond, and 3 pairs in a triple bond**

Lewis Structures



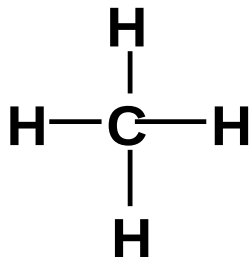
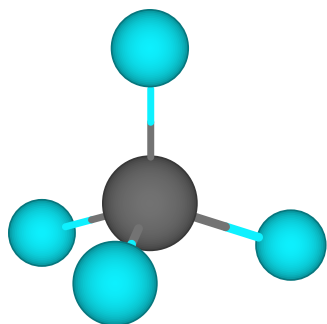
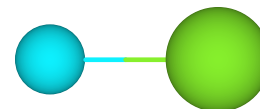
H_2O (8)

Water



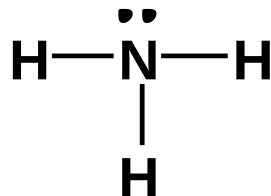
HCl (8)

Hydrogen chloride



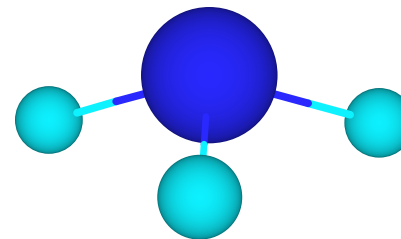
CH_4 (8)

Methane

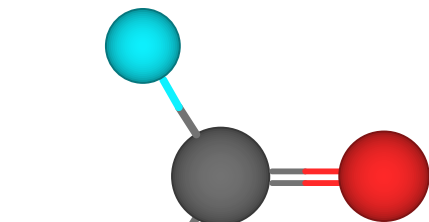


NH_3 (8)

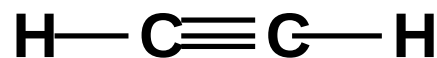
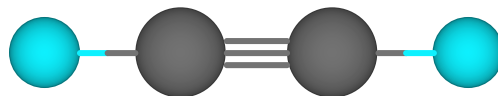
Ammonia



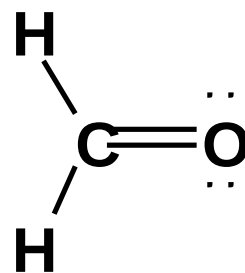
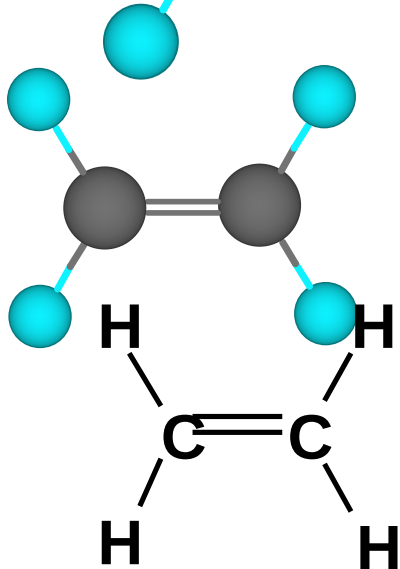
Lewis Structures



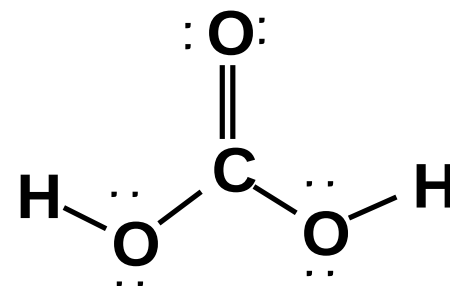
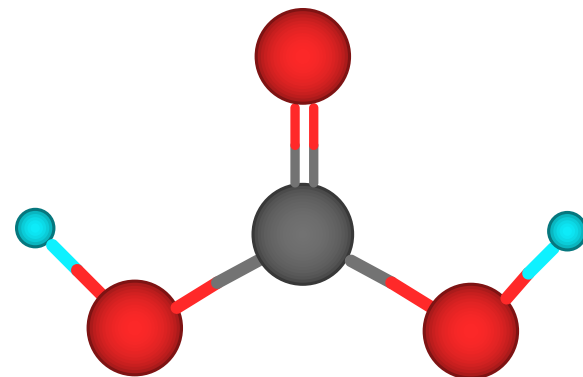
C_2H_4 (12)
Ethylene



C_2H_2 (10)
Acetylene



CH_2O (12)
Methanal



H_2CO_3 (24)
Carbonic acid

Lewis Structures

- In neutral molecules containing C, H, N, O, and halogen
 - hydrogen has one bond
 - carbon has 4 bonds and no unshared electrons
 - nitrogen has 3 bonds and 1 unshared pair of electrons
 - oxygen has 2 bonds and 2 unshared pairs of electrons
 - halogen has 1 bond and 3 unshared pairs of electrons

Formal Charge

- **Formal charge:** the charge on an atom in a molecule or polyatomic ion
 - write a Lewis structure for the molecule or ion
 - assign each atom all its unshared (nonbonding) electrons and one-half its shared (bonding) electrons
 - compare this number with the number of valence electrons in the neutral, unbonded atom
 - if the number is less than that assigned to the unbonded atom, the atom has a **formal positive charge**
 - if the number is greater, the atom has a **formal negative charge**

$$\text{Formal charge} = \begin{array}{c} \# \text{ of valence} \\ \text{electrons in} \\ \text{unbonded atom} \end{array} - \left(\begin{array}{cc} \text{all} & \text{one-half} \\ \text{unshared} & \text{of all shared} \\ \text{electrons} & \text{electrons} \end{array} \right)$$

Formal Charge

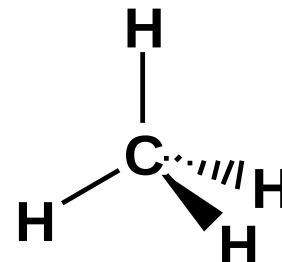
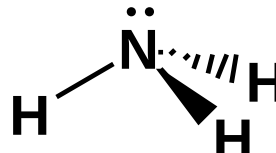
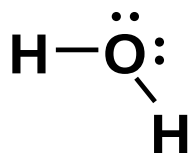
- **Exercise:** draw a Lewis structure and show all formal charges for these ions
- **Please submit your written answers by next class!**



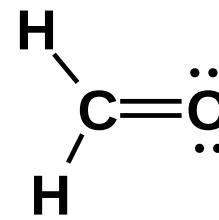
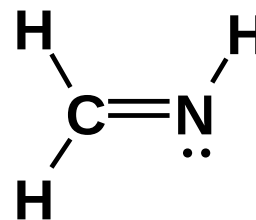
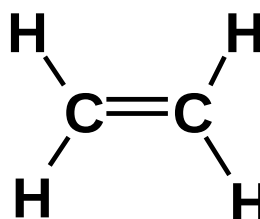
VSEPR Model

- This model is based on two concepts
 - atoms are surrounded by regions of electron density
 - regions of electron density repel each other

4 regions of e^- density
(tetrahedral, 109.5°)



3 regions of e^- density
(trigonal planar, 120°)

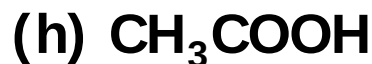
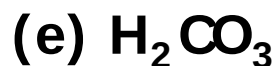


2 regions of e^- density
(linear, 180°)



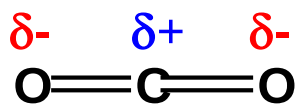
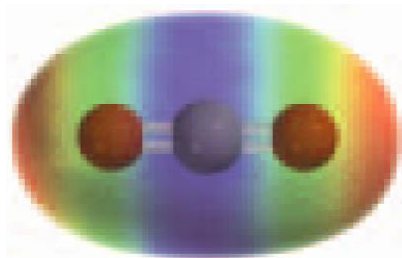
VSEPR Model

- **Exercise:** predict all bond angles for these molecules and ions
- **Please submit your written answers by next class!**

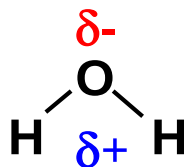


Polar and Nonpolar Molecules

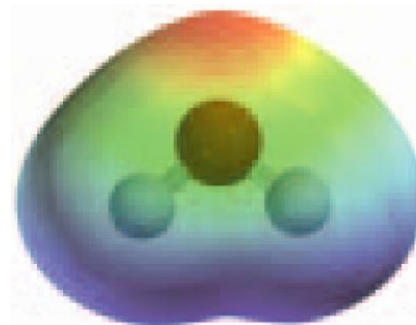
- A molecule is polar if
 - it has polar bonds and
 - its center of positive charge is at a different place than its center of negative charge



Carbon dioxide
(nonpolar)

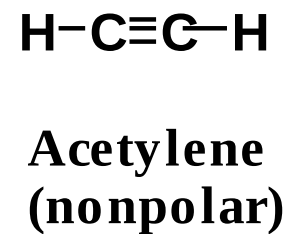
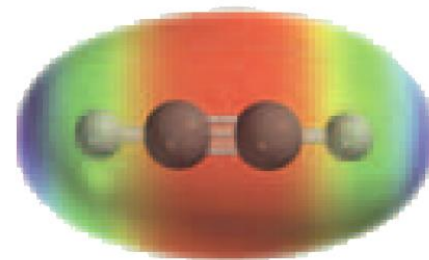
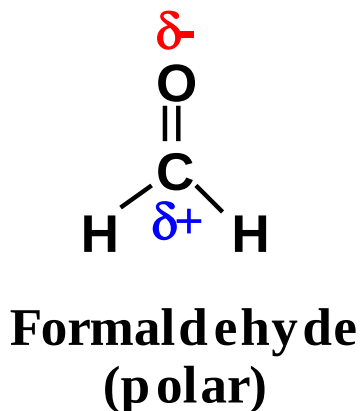
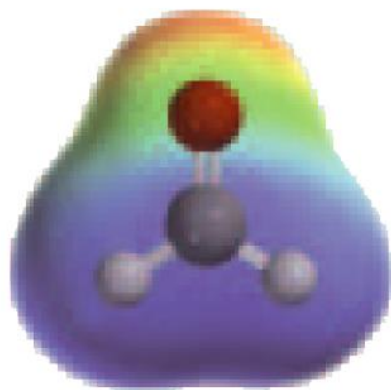
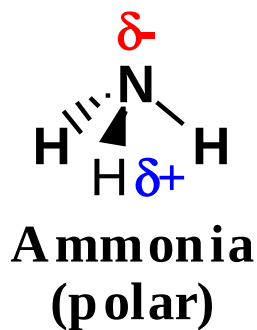
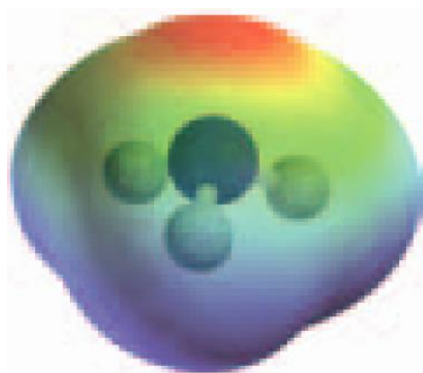


Water
(polar)



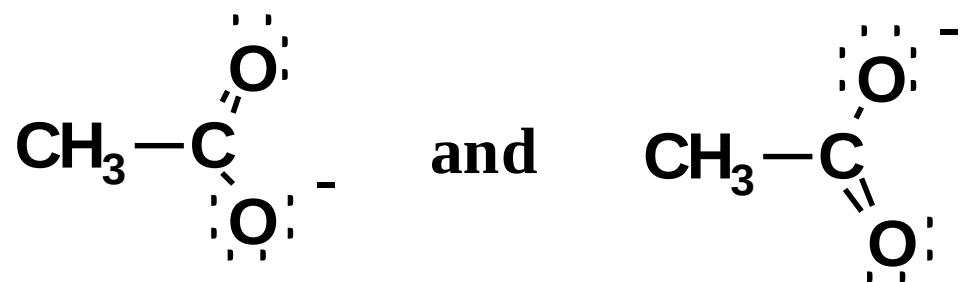
Polar and Nonpolar Molecules

- ammonia and formaldehyde are polar molecules
- acetylene is a nonpolar molecule



Resonance

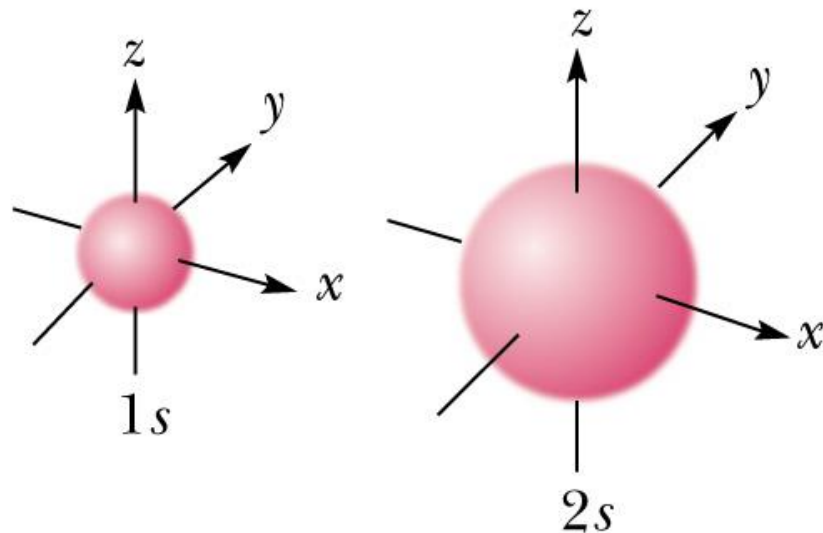
- For many molecules and ions, no single Lewis structure provides a truly accurate representation



Orbital Hybridizations

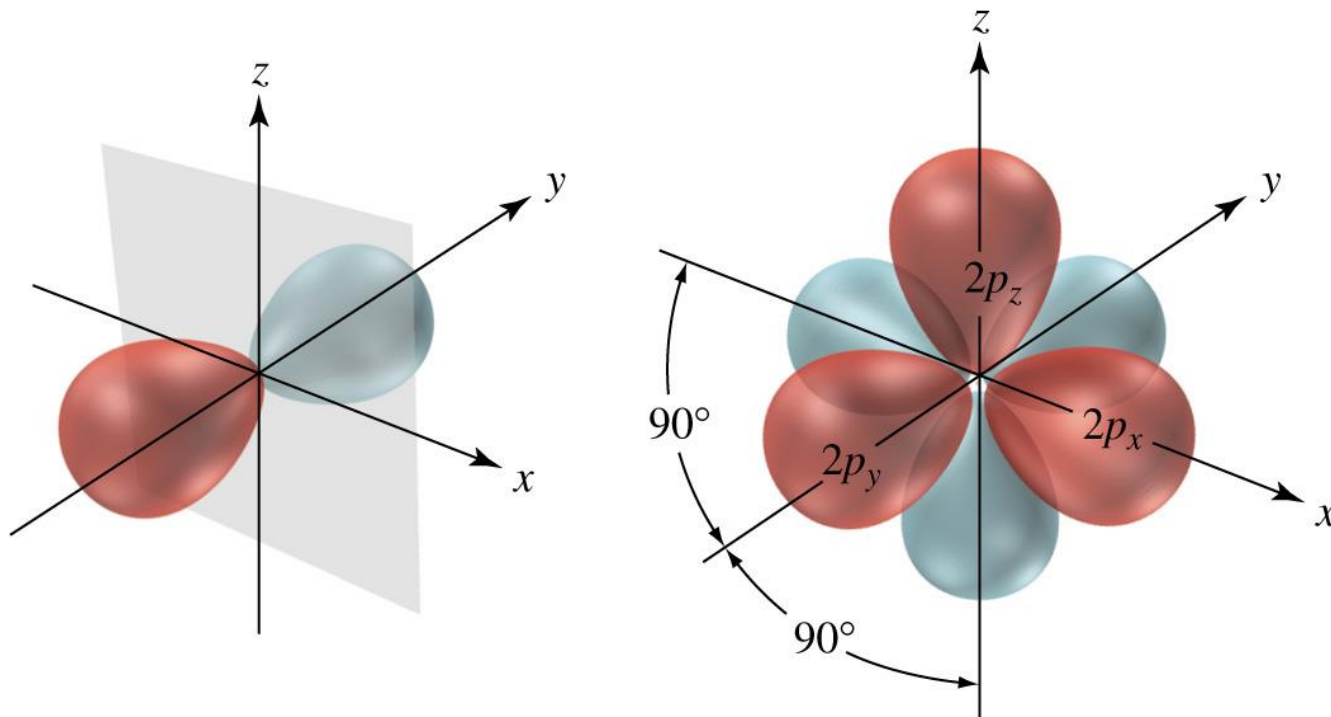
Shapes of Atomic Orbitals

- All s orbitals have the shape of a sphere, with its center at the nucleus
 - of the s orbitals, a 1s orbital is the smallest, a 2s orbital is larger, and a 3s orbital is larger still



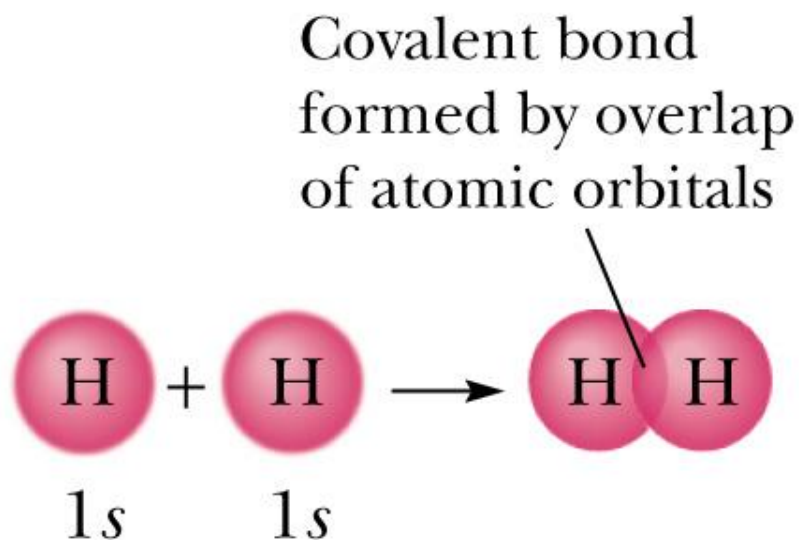
Shapes of Atomic Orbitals

- A *p* orbital consists of two lobes arranged in a straight line with the center at the nucleus



Orbital Overlap Model

- A covalent bond forms when a portion of an atomic orbital of one atom overlaps a portion of an atomic orbital of another atom
 - in forming the covalent bond in H-H, for example, there is overlap of the 1s orbitals of each hydrogen



Hybrid Orbitals

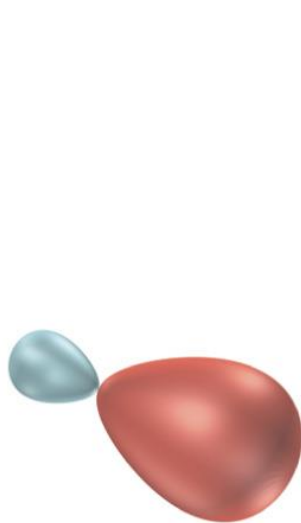
- **The Problem:**
 - overlap of 2s and 2p atomic orbitals would give bond angles of approximately 90°
 - instead we observe bond angles of approximately 109.5° , 120° , and 180°
- **A Solution**
 - hybridization of atomic orbitals
 - 2nd row elements use sp^3 , sp^2 , and sp hybrid orbitals for bonding

Hybrid Orbitals

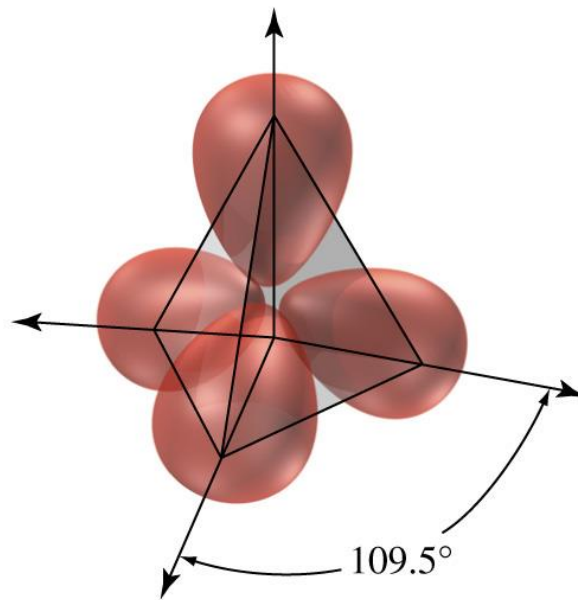
- We study three types of hybrid atomic orbitals
 - sp^3 (one s orbital + three p orbitals give four sp^3 orbitals)
 - sp^2 (one s orbital + two p orbitals give three sp^2 orbitals)
 - sp (one s orbital + one p orbital give two sp orbitals)
- Overlap of hybrid orbitals can form two types of bonds, depending on the geometry of the overlap
 - σ bonds are formed by “direct” overlap
 - π bonds are formed by “parallel” overlap

sp^3 Hybrid Orbitals

- Each sp^3 hybrid orbital has two lobes of unequal size
- The four sp^3 hybrid orbitals are directed toward the corners of a regular tetrahedron at angles of 109.5°



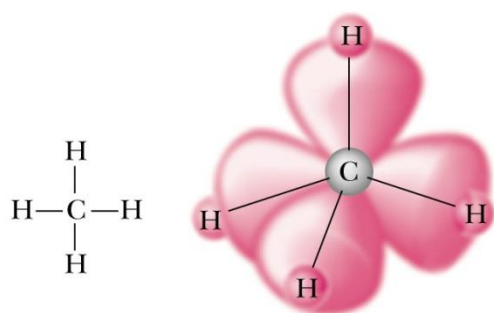
(a) An sp^3 orbital



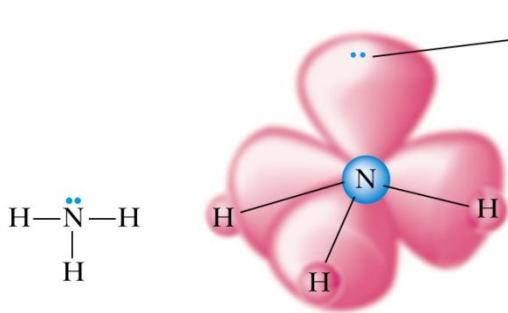
(b) Four tetrahedral sp^3 orbitals

sp^3 Hybrid Orbitals

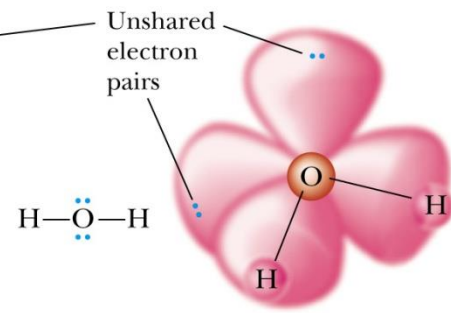
- orbital overlap bonding in water, ammonia, and methane



Orbital overlap model
Methane



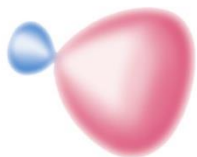
Orbital overlap model
Ammonia



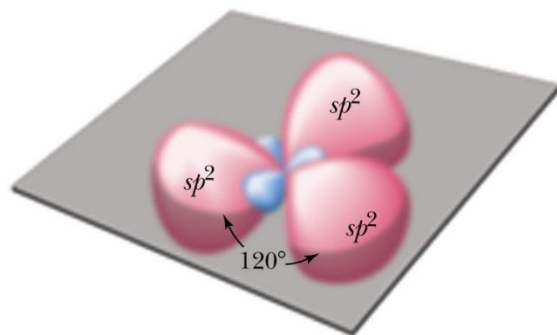
Orbital overlap model
Water

sp^2 Hybrid Orbitals

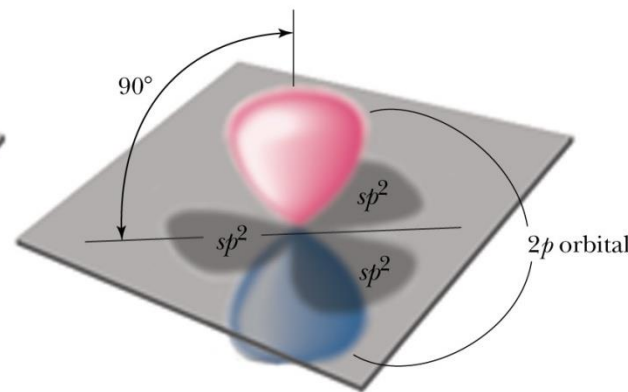
- An sp^2 hybrid orbital has two lobes of unequal size
 - the three sp^2 hybrid orbitals are directed toward the corners of an equilateral triangle at angles of 120°
 - the unhybridized $2p$ orbital is perpendicular to the plane of the three sp^2 hybrid orbitals



(a) An sp^2 orbital



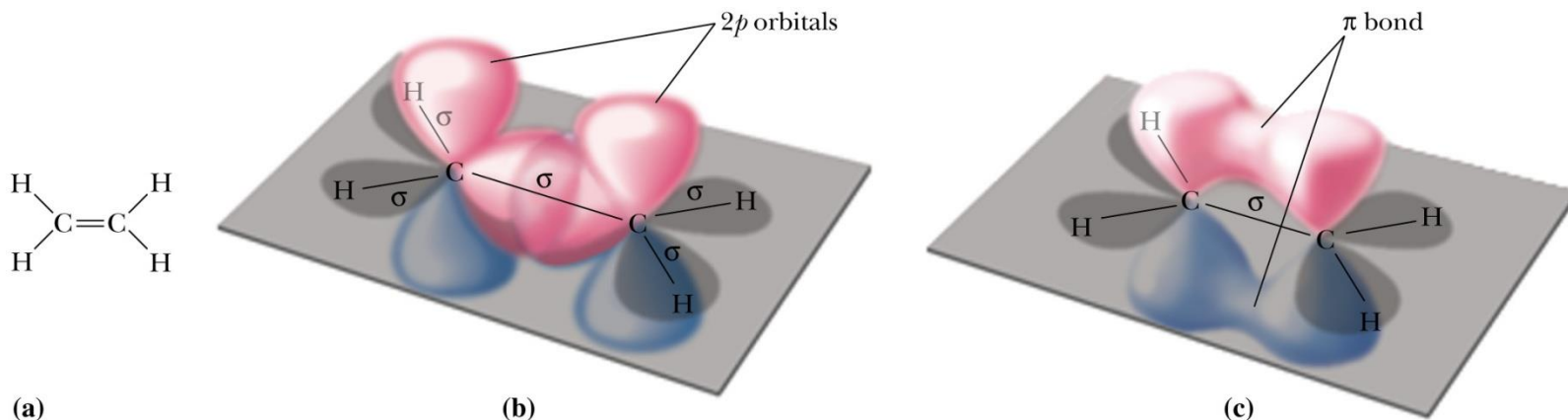
(b) Three sp^2 orbitals



(c) Three sp^2 orbitals and an unhybridized $2p$ orbital

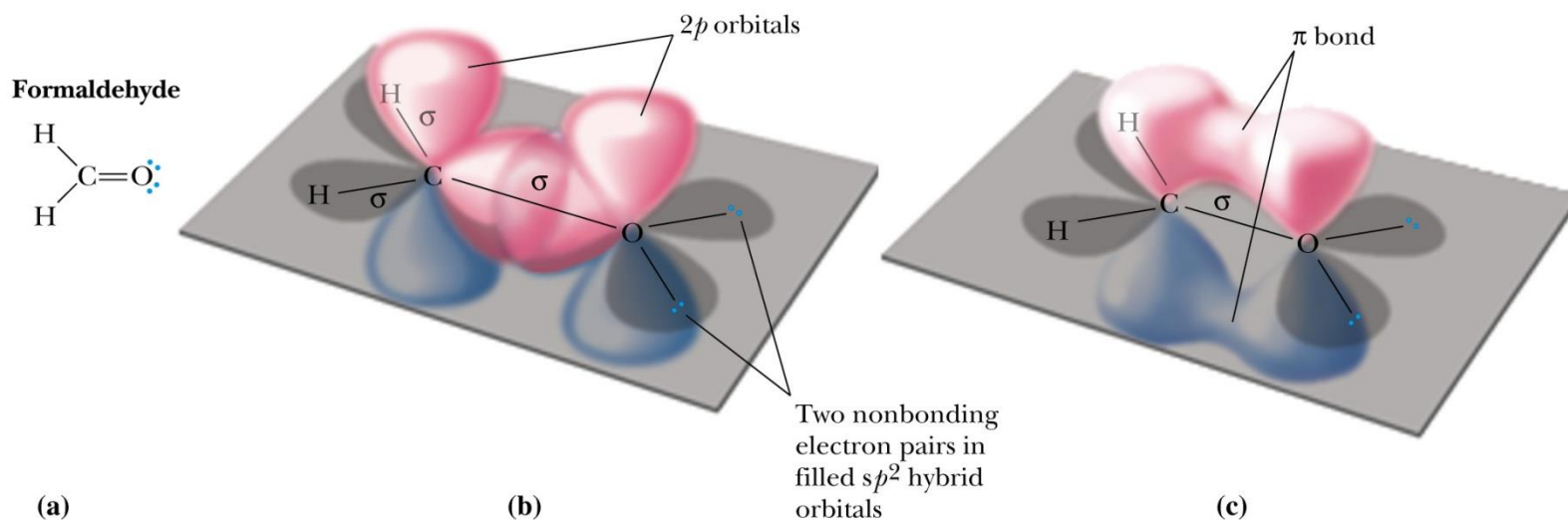
sp^2 Hybrid Orbitals

- a carbon-carbon double bond consists of one sigma (σ) bond and one pi (π) bond



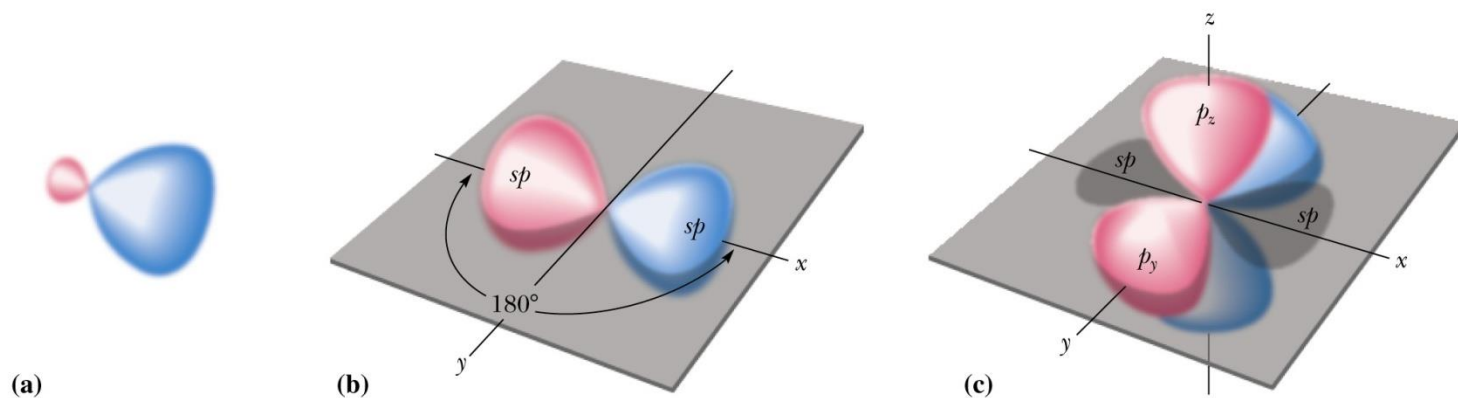
sp^2 Hybrid Orbitals

- a carbon-oxygen double bond also consists of one sigma (σ) bond and one pi (π) bond



sp Hybrid Orbitals

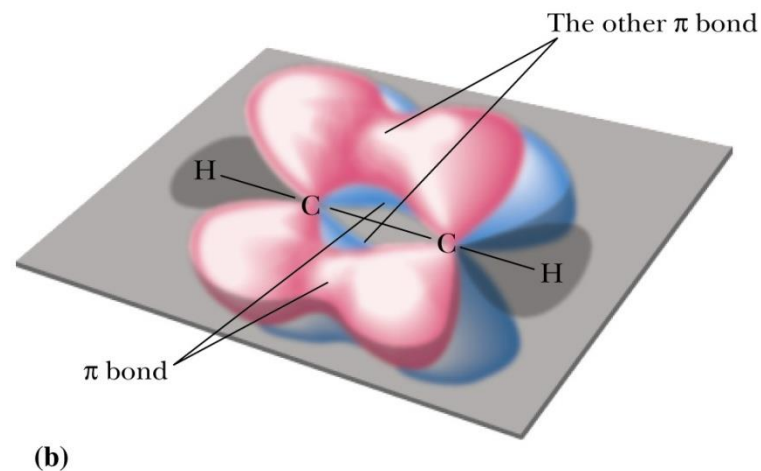
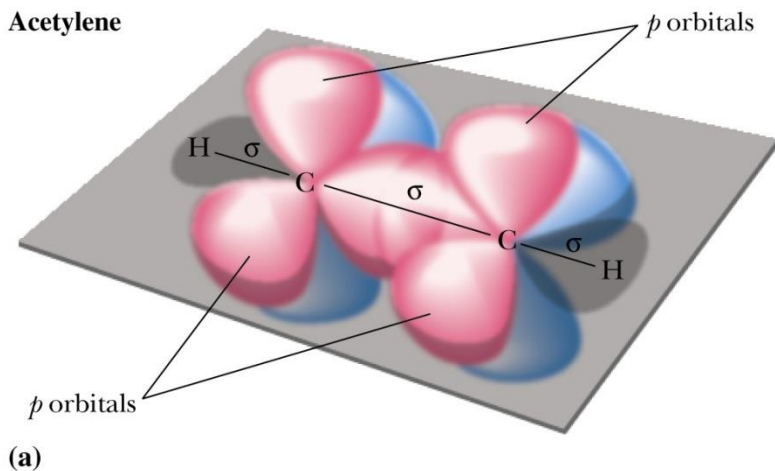
- Each *sp* hybrid orbital has two lobes of unequal size
 - the two *sp* hybrid orbitals lie in a line at an angle of 180°
 - the two unhybridized $2p$ orbitals are perpendicular to each other and to the line through the two *sp* hybrid orbitals



sp Hybrid Orbitals

- a carbon-carbon triple bond consists of one sigma (σ) bond and two pi (π) bonds

Acetylene



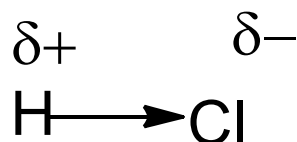
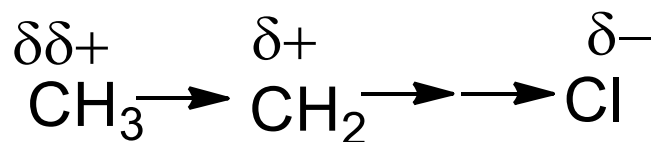
Hybrid Orbitals

- Summary of orbitals and bond types

Hybrid-ization	Types of Bonds to Carbon	Example	Name
sp^3	four sigma bonds	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{C}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array}$	Ethane
sp^2	three sigma bonds and one pi bond	$\begin{array}{c} \text{H} \quad \quad \text{H} \\ \diagdown \quad \diagup \\ \text{C}=\text{C} \\ \diagup \quad \diagdown \\ \text{H} \quad \quad \text{H} \end{array}$	Ethylene
sp	two sigma bonds and two pi bonds	$\text{H}-\text{C}\equiv\text{C}-\text{H}$	Acetylene

Factors Influencing Electron Availability and Reactivity Of Organic Compounds

- Inductive effect:** Electron donation or withdrawal through the sigma bonds of a molecule.



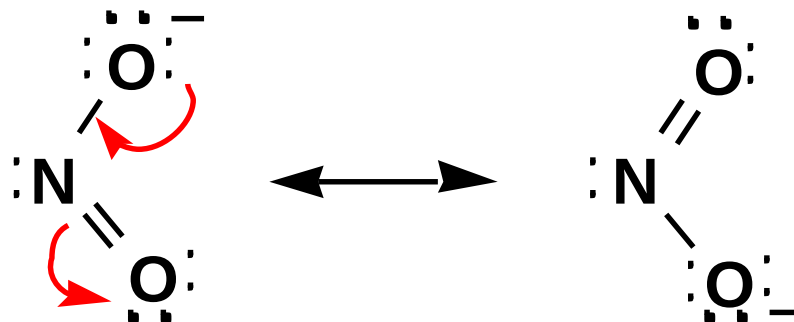
- Polarization of one bond caused by the polarization of an adjacent bond is the inductive effect.
- The effect is greatest for adjacent bonds but may also be felt farther away

-
-
- **Mesomeric or resonance effects**
 - **Electron donation or withdrawal through π bonds which can be represented specifically by drawing resonance forms**
 - Linus Pauling - 1930s
 - **many molecules and ions are best described by writing two or more Lewis structures**
 - **individual Lewis structures are called contributing structures**
 - **connect individual contributing structures by a double-headed (resonance) arrow**
 - **the molecule or ion is a hybrid of the various contributing structures**

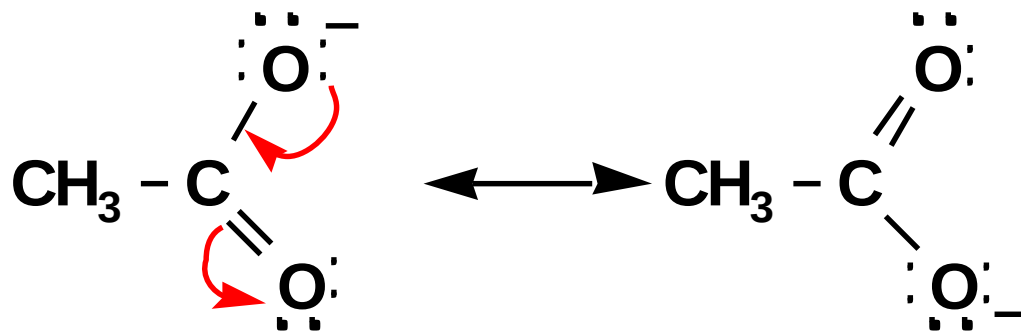
Resonance

- Examples

**Nitrite ion
(equivalent
contributing
structures)**



**Ethanoate ion
(equivalent
contributing
structures)**



Resonance

- **Curved arrow:** a symbol used to show the redistribution of valence electrons
- In using curved arrows, there are only two allowed types of electron redistribution:
 - from a bond to an adjacent atom
 - from an atom to an adjacent bond
- Electron pushing by the use of curved arrows is a survival skill in organic chemistry
 - **learn it well!**



Resonance

- **All acceptable contributing structures must:**
 - 1. have the same number of valence electrons**
 - 2. obey the rules of covalent bonding**
 - no more than 2 electrons in the valence shell of H
 - no more than 8 electrons in the valence shell of a 2nd period element (O, N, C etc)
 - 3rd period elements may have up to 12 electrons in their valence shells (S, P etc)
 - 3. differ only in distribution of valence electrons**
 - 4. have the same number of paired and unpaired electrons**

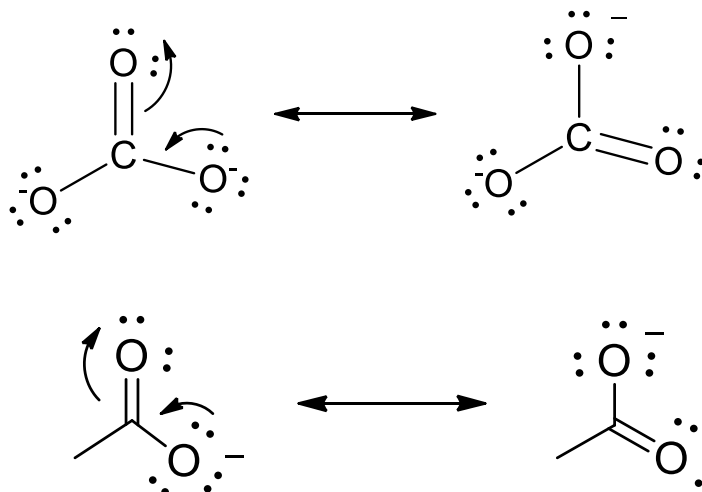
Resonance

- Examples of ions and a molecule best represented as resonance hybrids

carbonate ion CO_3^{2-}

acetate ion CH_3COO^-

acetone CH_3COCH_3

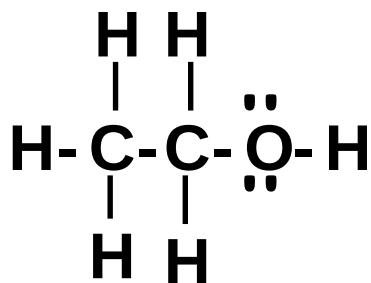


Functional Groups

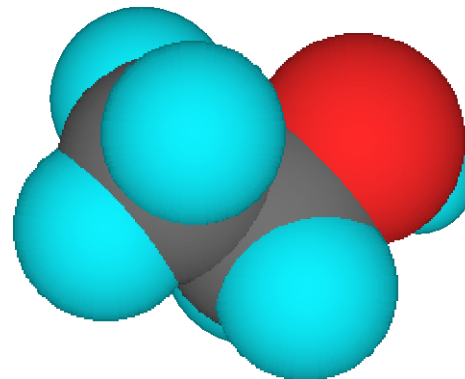
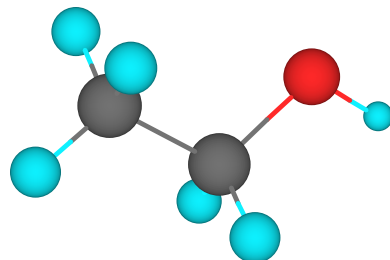
- **Functional group:** an atom or group of atoms within a molecule that shows a characteristic set of physical and chemical properties
- Functional groups are important for three reasons, they are
 - the units by which we divide organic compounds into classes
 - the sites of characteristic chemical reactions
 - the basis for naming organic compounds

Functional Groups

- Hydroxyl (OH) group of alcohols

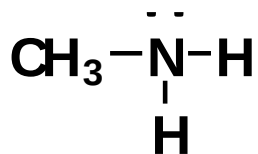
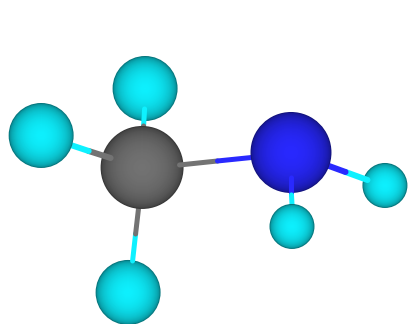


**An alcohol
(Ethanol)**

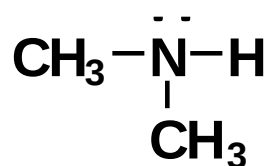
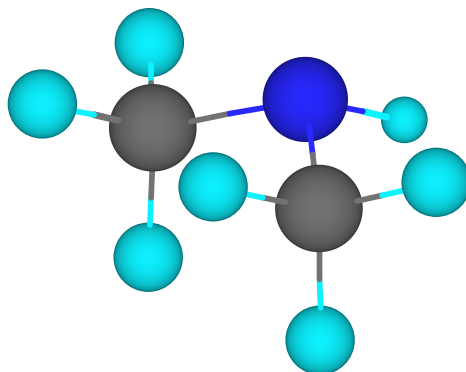


Functional Groups

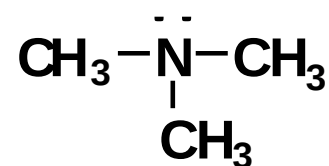
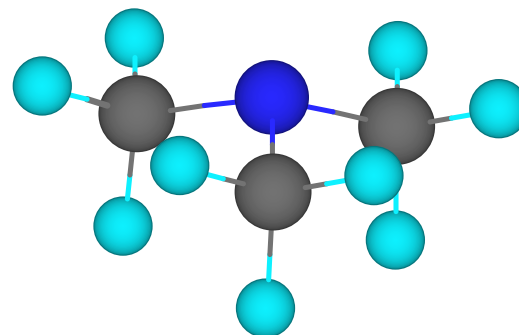
- Amino group of 1°, 2°, and 3° amines



(a 1° amine)



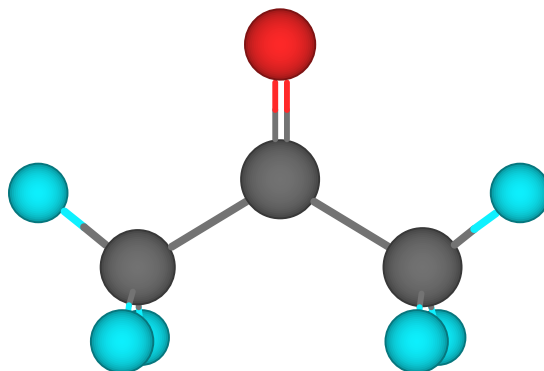
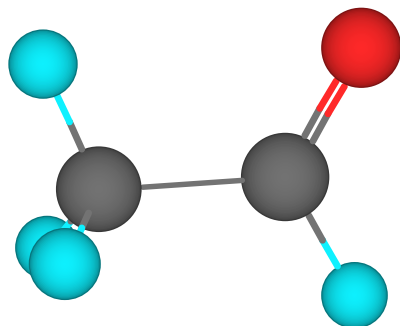
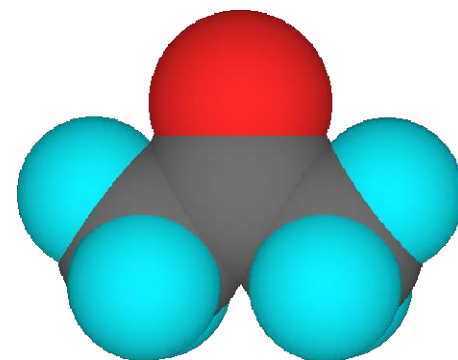
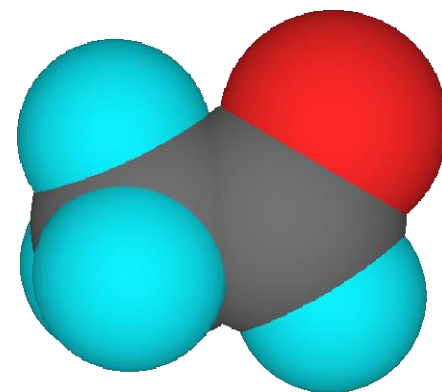
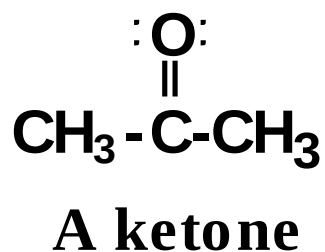
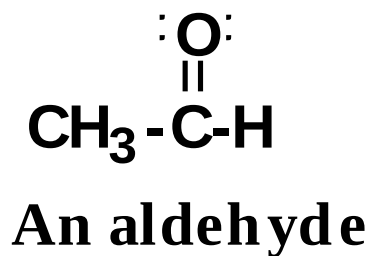
(a 2° amine)



(a 3° amine)

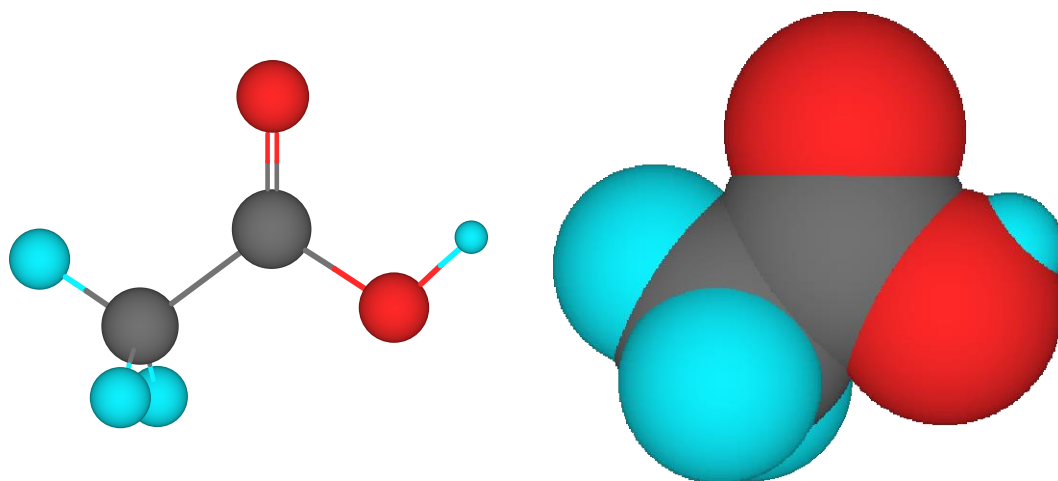
Functional Groups

- Carbonyl group of aldehydes and ketones

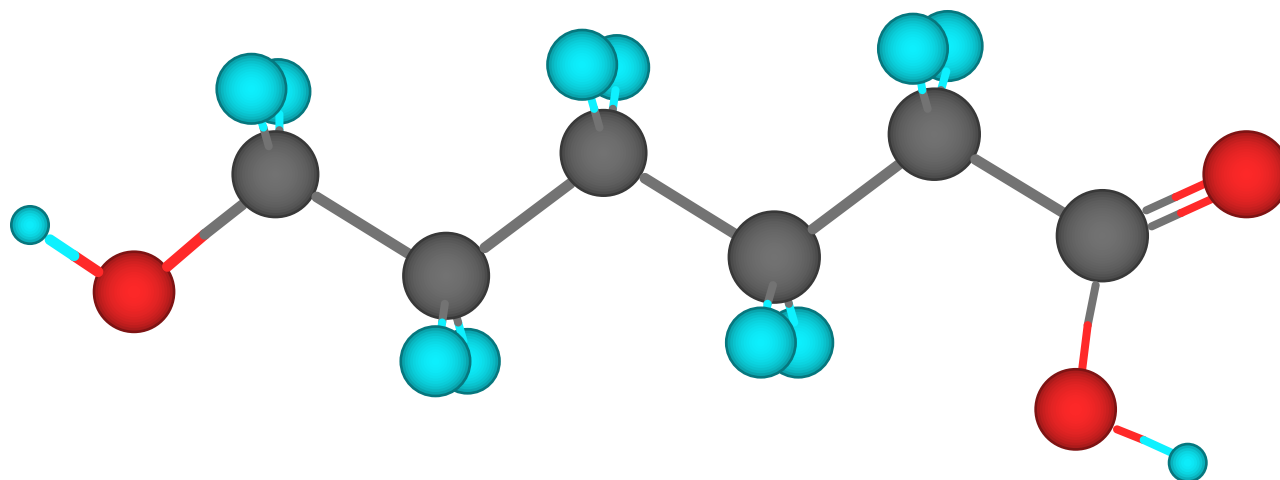


Functional Groups

- Carboxyl group of carboxylic acids



Covalent Bonding & Shapes of Molecules



End Chapter 1